



The role of TCNQ for surface and interface passivation in inverted perovskite solar cells

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Abstract

The noticeable growth in the power conversion efficiency of solution-processed organo-inorganic halide perovskite solar cells (OIHPSCs) incited the photovoltaic community to look for limitations that hurdle the commercialization process. The surface and interface defects between the perovskite and electron transport layers are among the main challenges that cause significant non-radiative recombination losses, thereby they result in poor performance and stability. In this work, tetracyanoquinodimethane (TCNQ), a strong electron acceptor molecule, is applied at the interface between the photoactive perovskite and [6,6]-phenyl C₆₁ butyric acid methyl ester (PCBM) layers to modify the interface, and enhance device performance and stability. Steady-state and time-resolved photoluminescence measurements were used to characterize the role of the TCNQ passivation in reducing non-radiative recombination of charge carriers. Current density versus voltage (J-V) measurements show improvement in devices open-circuit voltage (V_{oc}), short-circuit current density (J_{sc}), and fill factor (FF) for devices with TCNQ interface passivation, which is attributed to suppressed non-radiative recombination. In addition, a noticeable improvement in the device's stability was observed. This study reveals the dual role of TCNQ passivation in improving the photoelectric properties and stability of ambient air processed perovskite devices with the pin architecture.

Keywords Organo-inorganic halide perovskite solar cells · Interface passivation · Recombination · Stability · Power conversion efficiency

Introduction

The supply of renewable and green energy is the main hurdle to implementing sustainable development goals. Different renewable energy sources have been widely deployed to mitigate the emission of greenhouse gases. In this regard, the photovoltaic community has been working on different solar cell materials to provide clean, affordable, and sustainable energy to achieve zero greenhouse gas emissions. These days, even though silicon solar cells are relatively efficient, clean, and sustainable, their wide-spread implementation to satisfy the strong demands for renewable energy is limited due to their costly production process of these devices [1, 2]. As an alternative technology, thin film solar cells which provide the advantage of flexibility and comparative affordability have been developed. However, these devices have been criticized for their low power conversion efficiencies (PCE), scarcity of used materials, and toxicity when compared with single-crystal silicon

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solar cells [3]. On the other hand, emerging photovoltaics develop clean and low-cost solar cells using organic and dye materials, even though their PCE and long-term stability are not eye-catching yet. Nowadays, the conspicuous growth of the PCE of organic-inorganic halide perovskite solar cells (OIHP) towards the Shockley-Queisser limit (33%), has astonished the photovoltaic research community [4]. Currently, OIHPSCs achieved a certified record PCE of 26.7%, making it on par with single crystal (non-concentrator) Si solar cells [5]. Their attractive optical and electrical properties like high absorption coefficient [6], large charge carrier diffusion length [7], high charge carrier mobility [8], and tunable bandgap [9] are accounted for such improvement in a short time. These solution-processed polycrystalline devices possess grain boundaries, trap centers, or defects owing to the poor crystallinity, and dangling bonds, thereby different recombination processes take place at their surface or interface [10, 11]. These variables provide room for the interaction of extrinsic and intrinsic factors like moisture, oxygen, light, temperature, and ion migration which can induce defects in either the surface or at the interface of the photo-absorber perovskite layer [12]. Consequently, they limit the generation, extraction, and transportation of the charge carriers and may cause deterioration of the photovoltaic device performance under ambient conditions. Inverted (p-i-n) planar perovskite solar cells known to exhibit better long-term stability compared to their normal (n-i-p) structure, though their PCE lags behind conventional structures. The interface of electron transport layer (ETL) and the photo-absorber perovskite layer is found to be among the major determinants that impact photovoltaic characteristics of such devices [13]. Specifically, the open circuit voltage (V_{oc}) and fill factor (FF) of these devices have been depending on the quality of the interfaces of the charge transporting and OIHP layers. Therefore, interface engineering as the recombination impediment is commonly applied to treat charge trapping centers which cause the recombination of charge carriers [14–18]. Hence, different compounds have been introduced as a thin interlayer in between ETL and photoactive layers, thereby they showed significant improvement in short current density (J_{sc}), V_{oc} and FF [19–24].

In this work, surface and interface defect mitigation is accomplished using TCNQ as the interface passivator between OIHP and PCBM. The results from measurements of dark current, steady-state photoluminescence, time-resolved photoluminescence, and intensity-modulated photovoltage spectroscopy (IMVS) confirm that our passivation strategy helps to reduce non-radiative recombination. As a result, improvement in V_{oc} , J_{sc} , and FF of the fabricated OIHPSCs was achieved through interface passivation with TCNQ. This indicates that TCNQ contributes to

facilitate electron injection from the OIHP layer to PCBM by minimizing defect densities from the interfaces of photo-absorber perovskite and PCBM layers. Moreover, maximum-power-point tracking (MPPT) measurement results show that encapsulated devices with interface passivation exhibit improved operational stability in the ambient environment. Finally, in this finding we show that TCNQ has a good contestant for minimizing interfacial defects in inverted OIHPSCs.

Experimental

Materials

Indium tin oxide (ITO) coated glass substrates ($15 \Omega/\text{cm}^2$), lead iodide (PbI_2 , Sigma-Aldrich, 99%), lead chloride (PbCl_2 , Alfa Aesar, 99.9%), (PCBM, Solenne BV), aluminum (Al), Poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT: PSS), Clevios PH1000), Zonyl FS-300 fluorosurfactant (40% in H_2O , Fluka), dimethylformamide (DMF, anhydrous, Sigma Aldrich), dimethylsulfoxide (DMSO, Anal. R. VWR chemicals, 99.5%), chlorobenzene ($\text{C}_6\text{H}_5\text{Cl}$, 99.8%, Sigma Aldrich), chloroform (Anal. R. VWR chemicals), acetone (VWR Chemicals), isopropanol (VWR Chemicals), acetylacetone (AA, Sigma Aldrich $\geq 99\%$), and Hellmanex detergent (Hellma Analytics) is purchased from commercial chemical suppliers and used as received. In addition, methylamine (33 wt%, in absolute ethanol, Sigma Aldrich), hydroiodic acid (HI (aq), 57 wt%, Sigma Aldrich), diethyl ether (VWR Chemicals), and ethanol (absolute, MERCK) were used to synthesize and purify methylammonium iodide (MAI) according to a procedure reported in the literature [25]. TiO_x sol-gel is also synthesized based on the procedure reported by Park et al. [26].

Preparation of the Perovskite Solution

The solution of MAPbI_3 was prepared by mixing defined molar ratios of lead halides and methyl ammonium iodide in a 1:1:4 mol ratio of PbI_2 , PbCl_2 , and MAI, respectively [27]. Then, the mixture of the precursors was dissolved in DMF containing 10.0% acetyl acetone (AA) as the solvent additive, and was stirred overnight at about 45°C [28]. Finally, the prepared solution was ready for device fabrication after filtering using a $0.45 \mu\text{m}$ polytetrafluoroethylene (PTFE) filter.

Device fabrication

ITO-coated glass substrates were cleaned by using Hellmanex surfactant, deionized water, acetone, and isopropanol

(IPA), respectively for about 20 min of each step in an ultrasonic bath. The cleaned substrate was dried with nitrogen gas (N_2). The hole transport layer (HTL), PEDOT: PSS (Clevios PH1000) solution was prepared by adding 7% dimethyl sulfoxide (DMSO) and 0.7% Zonyl® FS-300 fluorosurfactant and it was mixed well to enhance the conductivity [29]. It was spin-coated at 1250 and 2000 rpm for 45 s and annealed at 120 °C for 15 min. The films were washed by IPA at 2000 rpm for 2 s and 4000 rpm for 10 s and annealed for 15 min at 120 °C. After cooling the PEDOT: PSS films to room temperature, the perovskite precursor solution ($MAPbI_3$), which was filtered using a 0.45 μm polytetrafluoroethylene (PTFE) filter was spin-coated at 1250 rpm for 17 s and 2000 rpm for 5 s on the top of the hole transport layer. After spin coating, the samples were kept for 1 min before being transferred to the hot plate and annealed for 30 min at 110 °C. Subsequently, different concentrations of TCNQ solution (0.125, 0.25, and 0.5 mg/mL) in chlorobenzene (CB) were prepared and spin-coated on top of the OIHP layer as interface layer at 5000 rpm for 30 s to find the optimal concentrations that yield best results. Then after, 20 mg/mL of PCBM solution in chlorobenzene and chloroform (1:1 volume ratio) was spin-coated on top of the TCNQ thin layer as electron transport layer (ETL) at 1250 rpm for 16 s and 2000 rpm for 15 s. Henceforth, TiO_x sol-gel was spin-coated on the top of the ETL at 5000 rpm for 30 s and annealed at 107 °C for 7 min as a buffer layer. All above procedures were carried out in an ambient environment. Finally, inverted OIHPSC with configuration: ITO/PEDOT: PSS/ $MAPbI_3$ /TCNQ/PCBM/ TiO_x /Al was completed by thermal evaporation of ~ 110 nm aluminum back contact with 0.16 cm^2 active area.

Characterization

Contact Angle Goniometer (Ossila, L2004A1) was used to carry out the contact angle measurements of the OIHP films at room temperature using 18 M Ω ultra-pure water. X-ray photoelectron spectroscopy (XPS) measurements were extracted by theta probe x-ray photoelectron spectrometer (Thermo Fisher). As the X-ray source, the system uses a monochromated Al-K α source with energy of 1486.6 eV. The spot diameter on the sample surface was 400 μm . For high-resolution scans, a pass energy of 20 eV and step size of 0.05 eV were used at the hemispherical analyzer. A dual flood gun which provides low-kinetic energy electrons and Ar^+ ions was used for charge compensation on the sample. Data acquisition and evaluation are performed via the Advantage software (Thermo Fisher). The smart background correction function in the Advantage software was used for the background correction of high-resolution XPS spectra, which is based on a modified Shirley background function.

To correct charging, the binding energies of all spectra were referenced to the values of the $Pb4f_{7/2}$ binding energy of Pb(0) artifacts at 136.5 eV [30]. The thickness of different layers was measured by the profilometer (Bruker Dektak, XT). The surface morphology of the films was characterized by using Bruker Innova atomic force microscopy (AFM) and scanning electron microscopy (SEM).

Absorbance, transmittance, and reflectance of $MAPbI_3$ thin films with and without the TCNQ interface passivation layer were recorded by using a UV-Vis-NIR spectrometer (Perkin Elmer Lambda 1050). Photoluminescence spectra of sample films were measured using a Shamrock SR-303i monochromator and an Andor iDus Si-CCD detector. Samples were excited at 488 nm with 5 mW power using a COHERENT OBIS 488–150 LS laser. A set of long-pass filters was used to avoid any distortion of the recorded spectra by the laser light. Time-resolved photoluminescence measurements were performed using a time-correlated single photon counting (TCSPC) measurement setup. The setup consists of a DeltaNu DNS-300 monochromator (slit widths: 3 mm), a Becker & Hickl SPC 150 TCSPC module, and a PMC-100-1 photomultiplier. As an excitation light source, the setup uses a NKT Photonics SuperK FIANIUM FIU-15 with a pulse picker, connected to a Photon Etc LLTF Contrast VIS monochromator. TCSPC measurements were performed at an excitation wavelength of 530 nm. To characterize the photovoltaic (PV) performance of the perovskite solar cells photocurrent density-photo voltage (J-V) response was measured by using the Keithley-2400-LV source meter. A LOT-QD solar simulator with a 1000 W xenon lamp providing AM1.5 global spectrum was used for irradiation. The intensity of the solar simulator was calibrated by the silicon reference diode. Similarly, the dark current density (J_D)-voltage response of hole-only devices was measured using a Keithley-2400-LV source meter to get the space-charge limited current (SCLC) curves. The external-quantum efficiency (EQE) was measured by using an optical setup consisting of a lock-in amplifier (SR830, Stanford Research Systems) and a Jaissle 1002 potentiostat functioning as a preamplifier. Devices were illuminated with light from a xenon lamp passing through a monochromator (Oriel Cornerstone). A filter wheel holding long-pass filters and a mechanical chopper were mounted between the xenon lamp and a monochromator with chopping frequencies in the range of 10–200 Hz was chosen. A calibrated silicon diode (Hamamatsu S2281) was used as a reference to measure the light intensity at each wavelength. Intensity-modulated photovoltage spectroscopy (IMVS) measurements were executed at V_{oc} conditions under illumination with 10% light-modulated amplitude to gain insight into the recombination dynamics in devices with and without TCNQ passivation. The parameters were set to measure the

perturbed V_{oc} which is DC-voltage with 10% light intensity perturbation in the frequency range from 1 MHz to 50 mHz by using the PhotoEchem software. Then, the values of LED light intensity were set by the program to vary the intensity by controlling the current applied to the LED source.

Results and discussions

Chemical Interaction

The lone pair electrons in water molecules create Lewis basicity that binds with Lewis acid, PbI_2 while the solution of the OIHP is prepared. In turn, it contributes to the improvement of crystal quality and surface roughness, similar to other Lewis base additives [31, 32]. However, the ingress of moisture through charge-transporting layers is considered to be the prominent problem that affects the stability of the organo-metal halide perovskite solar cells (OMHPSCs) [33]. In the ambient environment, moisturized air is among the major extrinsic factors that cause degradation of the OMHP layers [34, 35]. In this respect, buffering or hydrophobic layers either between the perovskite layer and charge transporting layer (CTL) or between CTL and electrodes have been applied as the efficient and effective approaches to mitigate negative impacts of moisture, O_2 , and ions on the perovskite layer [36–38]. In this work, we used tetracyanoquinodimethane (TCNQ) (Fig. 1c) as the interface passivation layer between OMHP layer and PCBM layer. Figure 1a shows the schematic structure of inverted (p-i-n) planar OIHPSCs fabricated with TCNQ passivation interlayer between the OIHP and phenyl-C61-butyric acid methyl ester (PCBM) layers. Matching energy level alignment of the passivation layer with the different layers is crucial to prevent trapping or non-radiative recombination, which negatively impacts the overall performance and long-term stability of the devices [39] (Fig. 1b). In this work, the lowest unoccupied molecular orbital (LUMO) of the TCNQ, which was reported as -4.23 V by K. Kanai et al. was applied [40]. Contact angle measurement was employed to assess the wetting of the OMHP thin films without and with TCNQ passivation layer (Fig. 1d and e). The results show that films treated with the TCNQ exhibit higher contact angle, implying a more hydrophobic surface compared to the control films. This indicates that the interface-passivated devices could possess enhanced long-term stability under ambient conditions compared to devices without TCNQ passivation layer. To gain detailed insight regarding the surface chemical properties of the TCNQ treated films with (elemental composition, the electronic states of Pb and I atoms, and their interaction with TCNQ), X-ray photon electron spectroscopy (XPS) characterization was used. Figure 1f and g

show the XPS spectra of the Pb4f and I3d core levels of perovskite films without and with TCNQ passivation layer. For both core levels, no shift of the spectra upon application of the TCNQ film can be observed, which suggests that no significant chemical interactions between TCNQ and Pb^{2+} or I^- occur at the interface. Fig. S 1(a-d) shows the XPS spectra of C1s, N1s and O1s of the OMHP thin films without and with TCNQ passivation. Particularly, new C1s and N1s peaks which were observed at 284.7 eV and 400.8 eV that can be attributed to aromatic C=C and -CN groups of the TCNQ molecule and indicate the presence of the passivation layer on the OMHP surface.

Topographic response

The role of TCNQ on the morphology or crystal grain size was scrutinized using SEM and AFM. SEM images of films without and with TCNQ are shown in Fig S 2 a & b. There is no significant impact of TCNQ molecule on the morphology or the crystal size of the photoactive OIHP films. Similarly, AFM images (Fig S 2 c & d) support the findings of the SEM data. The extracted AFM images suggest that TCNQ treated OIHP films were rougher than the reference.

Optoelectronic responses

The UV-vis-IR absorbance of the OIHP films with and without TCNQ coating were also studied. As shown in (Fig. S3), TCNQ treated OIHP film shows improvement in optical properties specifically in absorbance in UV-vis-IR region.

Each interface in OIHPSCs play pivotal role on the dynamics of the charge carrier transfer. Particularly, interfaces defects that are responsible for non-radiative charge carrier recombination have to be controlled in order to improve recombination rate and photovoltaic properties. To resolve such issues, surface and interface passivation approaches are being used to subdue recombination processes that take place at the surface or/and across the interfaces [13, 41, 42]. In addition, to evaluate the dynamics of photoexcited charge carriers, steady-state photoluminescence (PL) emission of the glass/ITO/PEDOT: PSS/OIHP/ and glass/ITO/PEDOT: PSS/OIHP/TCNQ thin-films were measured (Fig. 2a). The steady-state PL spectra disclose the significant improvement in intensity of radiative emission of light when TCNQ was applied on the photoactive perovskite surface. This suggests a passivation effect of TCNQ, which can be related to a reduction of the defect density on the surface of the perovskite layer.

Time-resolved photoluminescence (TRPL) was used to explore charge carrier lifetimes of OIHP thin films without and with interface passivation using TCNQ (Fig. 2b; Table 1). The decay curves are fitted using a bi-exponential

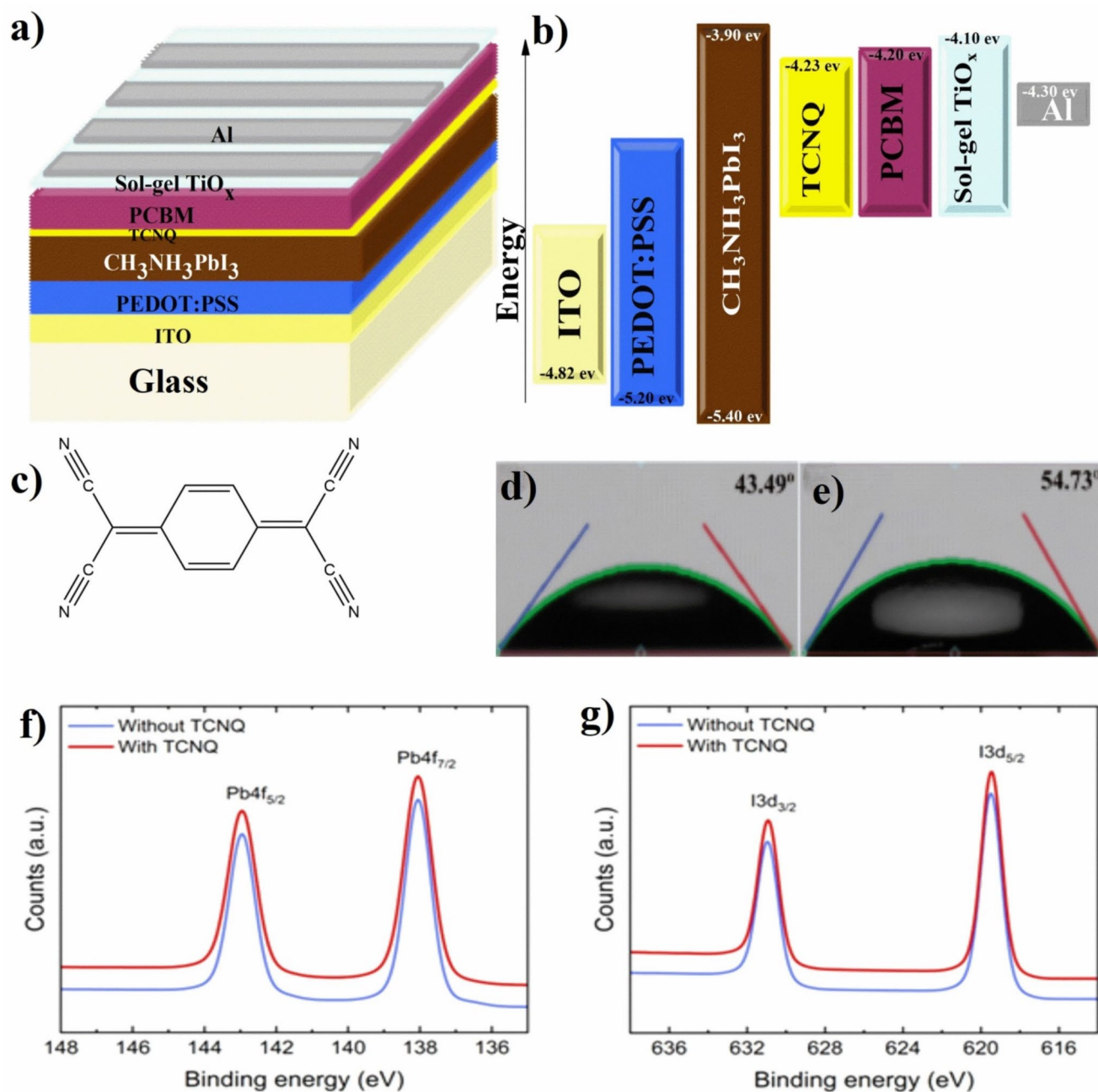


Fig. 1 (a) Typical schematic structure of inverted (p-i-n) OIHPSCs with the TCNQ passivation interlayer used in this work and, (b) the energy level alignment of the individual layers. (c) Chemical structure of TCNQ. Contact angle of water on the surface of OMHP (d) with-

out and (e) with TCNQ, respectively. Fitted XPS peaks (envelope) of (f) Pb4f and (g) I3d core levels of OMHP without and with TCNQ, respectively

decay function of the form $(x(t)=A_1\exp(-t/\tau_1)+A_2\exp(-t/\tau_2)+x_0)$, where τ_1 and τ_2 are the decay time constants, whereas A_1 and A_2 are the corresponding decay amplitudes and x_0 is a constant for the base-line offset [43]. In both the fast decay region (τ_1) and the slow decay region (τ_2), the TCNQ passivated OIHP films show a significantly increased decay time. This indicates a larger charge carrier lifetime in the photoactive layer that can be related to

reduced Shockley-Read-Hall charge recombination or surface recombination processes and further corroborates the trap or defect state passivation.

Among the electrical properties of the solar cell, the leakage current that is associated with shunt resistance loss and recombination at lower voltage helps to study electrical properties of semiconductors. It is reported that large leakage currents cause loss in the FF as well as the V_{oc} [44]. In

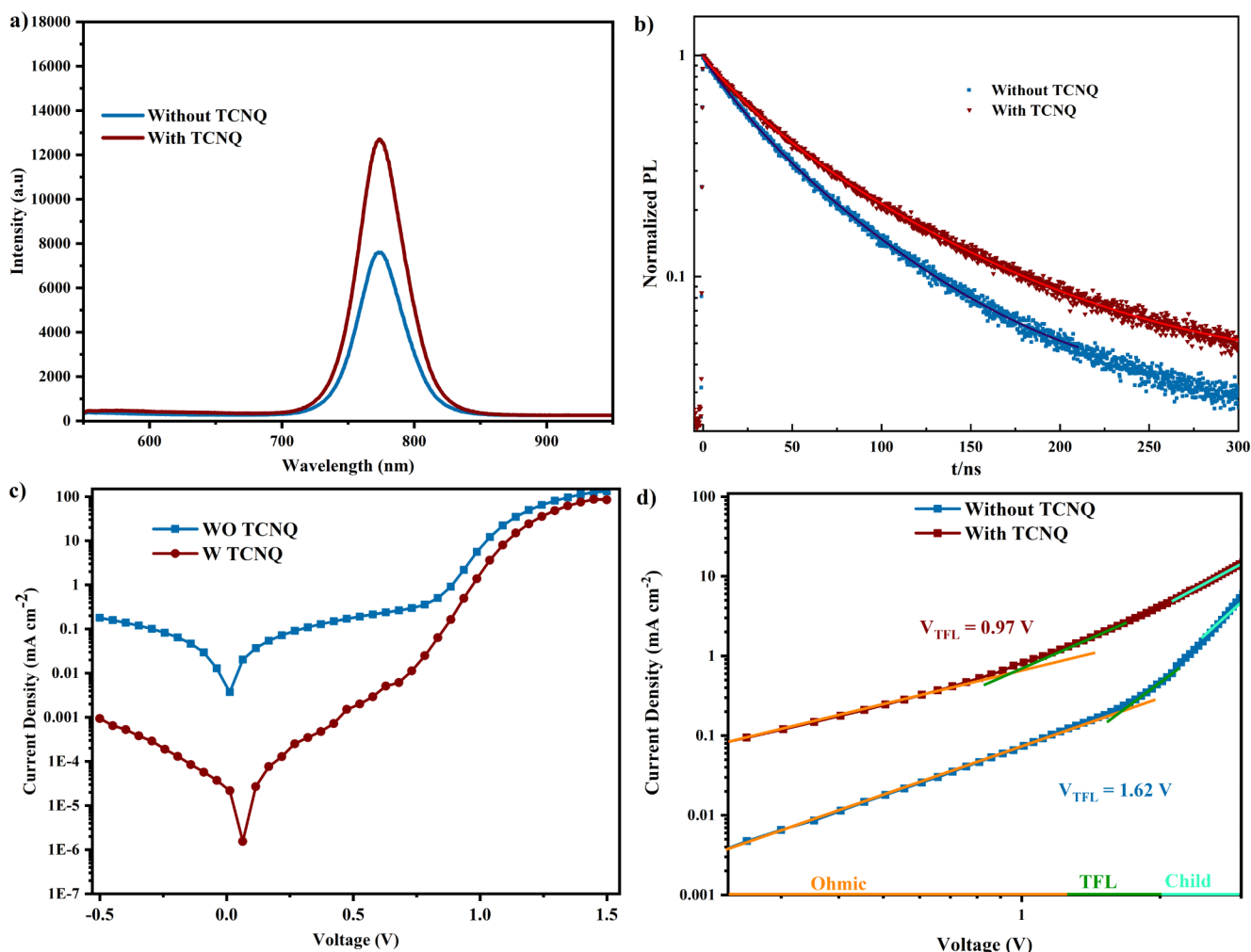


Fig. 2 Photoluminescence and electronic responses of $\text{CH}_3\text{NH}_3\text{PbI}_3$ perovskite with and without TCNQ passivation. **(a)** Steady-state photoluminescence (PL) and **(b)** time-resolved photoluminescence (TRPL) spectra of perovskite thin films on glass/ITO/PEDOT: PSS with and without TCNQ passivation (0.25 mg mL^{-1}), **(c)** semi-log plot

Table 1 Summary of the photoluminescence decay of the photoactive layer with and without TCNQ

Sample	A_1	A_2	τ_1 (ns)	τ_2 (ns)	τ_{avg} (ns)
Without TCNQ	0.34	0.61	20.8	60.2	46.2
With TCNQ	0.36	0.59	25.3	80.0	59.4

this regard, we measured dark current-voltage characteristics to investigate the role of TCNQ interface on minimizing leakage currents (Fig. 2c). Devices treated with TCNQ exhibits lower leakage current, attributing to the minimized charge carrier recombination at the interface; thereby, it leads to increased V_{oc} and FF.

Defect densities across the interface are also assessed for hole only devices with glass/ITO/PEDOT: PSS/ $\text{CH}_3\text{NH}_3\text{PbI}_3$ /PTAA/Al and Glass/ITO/PEDOT: PSS/ $\text{CH}_3\text{NH}_3\text{PbI}_3$ /TCNQ/PTAA/Al device structures using the space charge limited current (SCLC) method. The graphs

of dark current density-voltage characteristics of solar cell devices, and **(d)** space-charge-limited current (SCLC) of hole only device without and with TCNQ treatment. (blue=without TCNQ and red=with TCNQ)

in Fig. 2d show three different regimes, corresponding to ohmic, trap filled limited and trap free region. The density of defects or traps is directly related to the trap filled limited voltage (V_{TFL}) which is calculated using equation, $n_{\text{trap}} = 2\epsilon_0\epsilon V_{\text{TFL}}/qL^2$, where ϵ_0 is the vacuum permittivity constant, ϵ is the relative dielectric constant of MAPbI_3 , q is elementary charge and L is the thickness of the photo-absorber perovskite layer. Devices without ($V_{\text{TFL}} = 1.62 \text{ V}$, $n_{\text{trap}} = 2.3 \times 10^{16} \text{ cm}^{-3}$) and with TCNQ interface passivation ($V_{\text{TFL}} = 0.97 \text{ V}$, $n_{\text{trap}} = 1.4 \times 10^{16} \text{ cm}^{-3}$) are obtained. The result indicates that treatment of the perovskite layer with TCNQ suppresses the charge carrier trapping centers, and this improves charge carrier extraction through the interface.

Photovoltaic (PV) responses characterization

The role of TCNQ treatment on the photovoltaic performance of the prepared OIHPSCs was studied using the current density-voltage (J-V) characteristics which were measured under AM 1.5 G (100 mW cm^{-2}) illumination. Figure 3a show J-V characteristics of the reference, and sample devices (devices treated with TCNQ solutions having different concentration of, 0.125, 0.25 and 0.5 mg mL^{-1}), indicating that devices treated with 0.25 mg mL^{-1} of TCNQ display the optimum concentration. Figure 3b presents the J-V response (for the reverse scan) of the best-performing device treated with the optimum concentration (0.25 mg mL^{-1}) of TCNQ, exhibiting PCE of 16.5%, V_{oc} of $\sim 1.03 \text{ V}$, J_{sc} of $\sim 20.1 \text{ mA cm}^{-2}$, and FF of 79.4% compared to the reference device with PCE of 14.6%, V_{oc} of $\sim 0.98 \text{ V}$, J_{sc} of $\sim 19.2 \text{ mA cm}^{-2}$, and FF of about 77.5%. Moreover, the statistical distribution of the PV parameters (V_{oc} , J_{sc} , FF and PCE) of the 39 reference devices and more than 39 interface passivated OIHPSCs are plotted to indicate the impact of the passivation (Fig. 3(c-f)) on the photovoltaic performance. Comparatively, all interface passivated devices show improved performance compared to the reference devices. The average and best statistical data of reference and each TCNQ concentration of OIHPSCs were also tabulated in Table 2. In addition, samples with TCNQ passivation show that the impact of the hysteresis is insignificant on the performance of the device (Fig. S4). The finding suggests that the interface passivation augments the charge injection from the surface of the photoabsorber perovskite layer to the PCBM, as shown by the significant improvement of V_{oc} , J_{sc} and FF.

We measured the external quantum efficiency (EQE) spectra of the reference and interface-passivated OIHPSCs to investigate the role of TCNQ on the efficiency of charge injection, charge collection, and light harvesting. The result shows a significant increase in measured EQE in the wavelength range from 425 to 767 nm and in the integrated J_{sc} for passivated devices (Fig. 3g). This proves that TCNQ passivation contributes to reducing defect density and non-radiative recombination across the interface of the OIHP and PCBM, thus enhancing the performance of sample devices compared to the reference. Moreover, the operational stability of OIHPSCs without and with TCNQ interfacing was measured using the maximum power point tracking (MPPT) technique under continuous photoirradiation (100 mW cm^{-2} , AM1.5G). Figure 3h shows that solar cells with TCNQ passivation exhibit improved operational stability compared to the reference devices. In addition, the MPPT data of J-V characteristics, J_{MPP} and P_{MPP} were extracted and shown in Fig. S 4 (a, c & d). These data proves that TCNQ treated OIHP layers have significant influence in

improving thermal stability of the device. But its impact on the thermal stability of the voltage of the device is insignificant (Fig. S 4 b). In general, the finding proves that applying TCNQ as the surface and interface passivation plays a dual role in improving the photovoltaic performances as well as the operational stability of OIHPSCs.

Charge Carrier Recombination dynamics

Intensity-modulated photovoltage spectroscopy (IMVS) is applied to analyze the charge carrier recombination dynamics of OIHPSCs without and with TCNQ interface passivation. The Nyquist plot (Fig. 4a) shows two semicircle arcs in lower and higher transfer functions that are associated with bulk and interface recombination processes, respectively. The characteristic high-frequency semicircle arc of IMVS responses that is allied with a parallel combination of the recombination resistance of devices exhibits higher recombination resistance for devices with TCNQ interlayer passivation [45]. The imaginary transfer function of IMVS versus frequency is also plotted (Fig. 4b) to determine the critical frequency, which is related to the minimum current of the imaginary part of IMVS. The corresponding recombination lifetimes of electrons for the minimum frequency are calculated using $f_c = 1/2\pi\tau_c$ equation [46]. Thus, devices that are treated with TCNQ show a longer recombination lifetime, which is attributed to the lowered, interface recombination. On the other hand, open-circuit voltage as a function of light intensity displays a similar trend for control and sample devices (Fig. 4c). Devices treated with TCNQ show higher V_{oc} than the control OIHPSCs at a given light intensity (I). The light intensity dependence of devices without and with TCNQ interface passivation shows a similar trend with ideality factors (n_i) of 1.39 and 1.37, respectively. Devices treated with TCNQ show a smaller n_i value, which is in good agreement with the observed higher V_{oc} at a given light intensity and photovoltaic properties. Moreover, the recombination lifetime constants that are extracted from IMVS provide vital information regarding the degree of recombination of the charge carriers under V_{oc} . Our result shows that devices with TCNQ passivation exhibit a longer recombination life-time constant compared to reference devices under a given V_{oc} condition (Fig. 4d). In general, this finding reveals that passivation of interfaces is an efficient and effective approach to mitigating interfaces recombination issues.

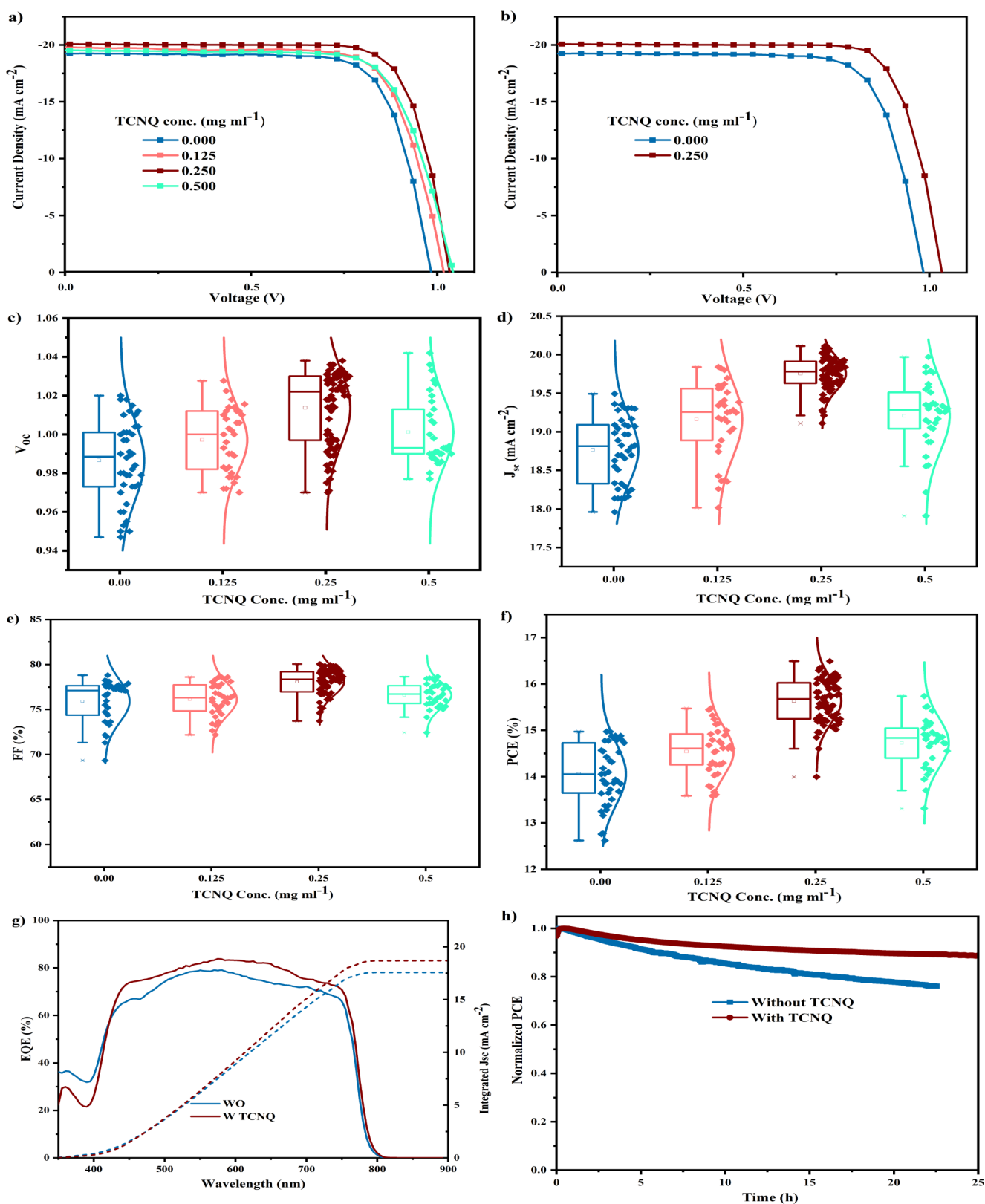
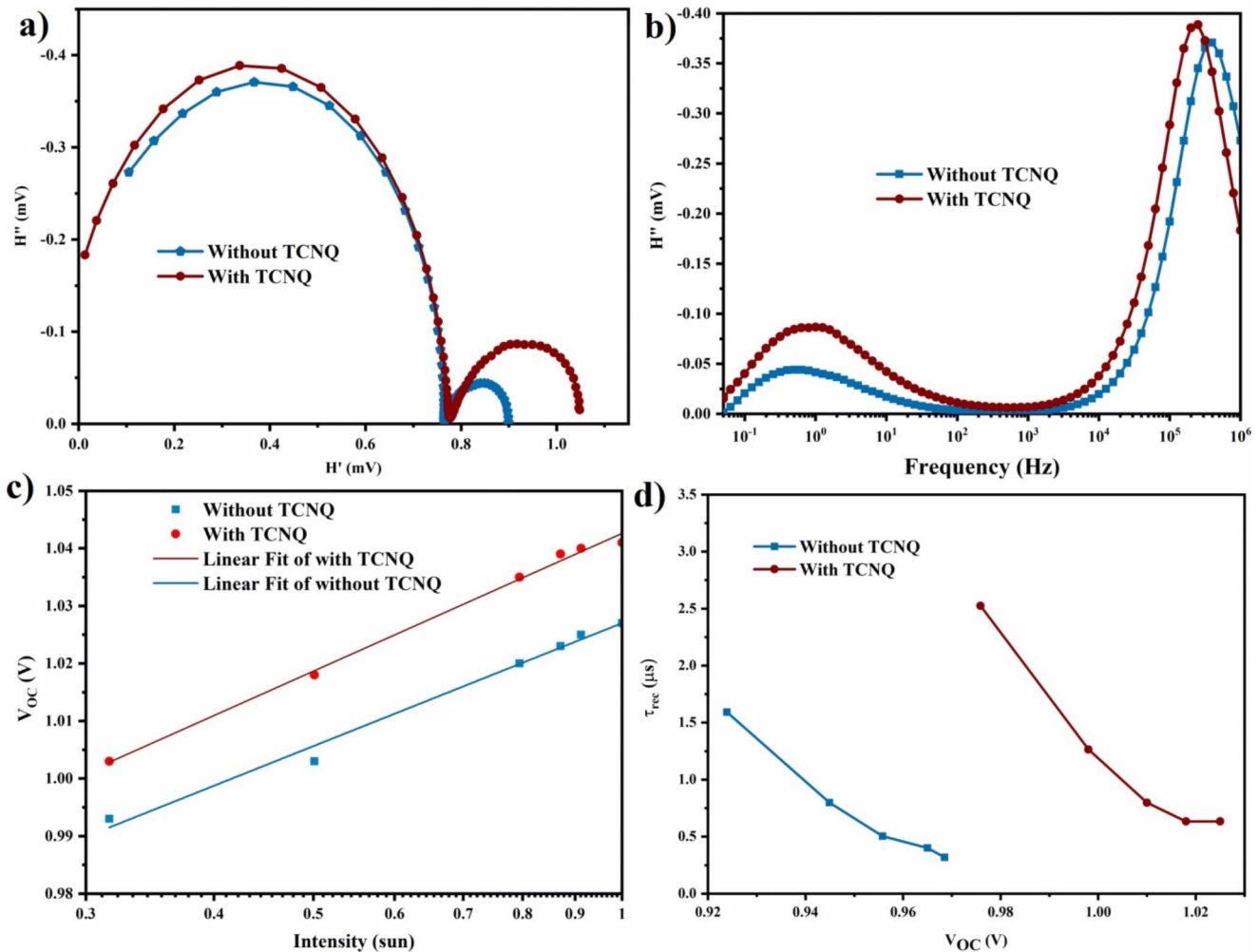


Fig. 3 PV properties of perovskite devices with and without TCNQ passivation. **(a)** J-V curves of devices treated with 0.0, 0.125, 0.25 and 0.5 mg mL⁻¹ of TCNQ. **(b)** J-V curves of perovskite solar cells treated with 0.0 (reference) and 0.25 mg mL⁻¹ (the optimal concentration) TCNQ. **(c, d, e, & f)** Statistical distribution of V_{oc} , J_{sc} , FF and PCE of

perovskite devices treated with 0.0, 0.125, 0.25 and 0.5 mg/mL TCNQ, respectively. **(g)** External quantum efficiency (EQE) and integrated J_{sc} , and **(h)** operational stability of encapsulated perovskite devices with and without TCNQ treatment obtained from MPP tracking measurements under 1 sun (100 mW cm⁻²) illumination

Table 2 Summary of PV parameters collected from J-V characteristics of perovskite devices treated with TCNQ interface passivator of various concentrations

Concentration (mg mL ⁻¹)		V _{oc} (V)	J _{sc} (mA cm ⁻²)	FF (%)	PCE (%)
0.00	Average	0.98±0.02	18.7±0.45	74.9±3.42	13.7±0.68
	Best	0.98	19.2	77.5	14.6
0.125	Average	0.99±0.01	19.1±0.57	76.2±1.85	14.5±0.56
	Best	1.02	19.8	76.8	15.5
0.25	Average	1.01±0.02	19.8±0.30	77.8±1.68	15.6±0.45
	Best	1.03	20.1	79.4	16.5
0.5	Average	1.00±0.02	19.2±0.54	76.6±1.43	14.7±0.57
	Best	1.04	19.8	75.2	15.5

**Fig. 4** Intensity-modulated photovoltage characteristics of MAPbI₃ perovskite solar cells containing 0.0, and 0.25 (mg/mL) of TCNQ. (a) Nyquist plots, (b) Imaginary transfer function (H'') versus frequency plots, (c) open-circuit voltage (V_{oc}) as a function of photon flux (cm⁻²s⁻¹) intensity, and (d) charge carrier recombination time constant (τ_{IMVS}) as a function of the V_{oc} of OIHPSCs

Conclusions

In this study, we show that employing TCNQ as an interface passivator between OIHP and PCBM layers can contribute to decreasing non-radiative recombination across the interface. This facilitates the injection of charge carriers and their mobility across the interface. It also enhances the hydrophobicity of the light absorber layer, which could contribute

to preventing moisture ingress to the perovskite layer from the ETL side. Furthermore, the interface-passivated devices processed in an ambient environment displayed improved PV characteristics. Devices treated with TCNQ passivation exhibit enhanced V_{oc}, J_{sc}, FF, and stability compared to control devices. This finding shows that the application of TCNQ as an interface passivation is an alternative approach

to reducing trap-assisted non-radiative recombination and improving the overall performance of perovskite devices.

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Declarations

Conflict of interest All authors declare no conflict of interest.

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